

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 2 Winter 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1000ES206A - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The unit of electrical resistance is:
a) Ohm b) Watt c) Henry d) Farad
- 2 Which element stores energy in the form of an electric field?
a) Resistor b) Inductor c) Capacitor d) Transformer
- 3 RMS value of a sinusoidal current is equal to:
a) Peak value b) Peak value / $\sqrt{2}$ c) Average value d) Zero
- 4 Which machine converts mechanical energy into electrical energy?
a) Motor b) Transformer c) Generator d) Induction motor
- 5 The working principle of an induction motor is based on:
a) Electromagnetic induction b) Electrostatics
c) Ohm's law d) Coulomb's law
- 6 CRO is mainly used to measure:
a) Resistance b) Power
c) Frequency and waveform d) Energy
- 7 Moving coil instruments are:
a) AC instruments b) DC instruments
c) Both AC & DC d) None of these
- 8 Digital Storage Oscilloscope is mainly used to:
a) Measure DC voltage only b) Display waveform and frequency
c) Measure current only d) Measure resistance
- 9 The total resistance of three resistors 2 Ω , 3 Ω , and 6 Ω connected in series is:
a) 11 Ω b) 1 Ω c) 6 Ω d) 3 Ω
- 10 Kirchhoff's Current Law (KCL) is based on:
a) Conservation of energy b) Conservation of charge
c) Ohm's law d) Lenz's law
- 11 Power factor in a purely resistive AC circuit is:
a) 0 b) 0.5 c) 1 d) -1

- 12 A Series DC motor is suitable for:
 a) Constant speed applications b) High starting torque applications
 c) Precision speed control d) Low power applications
- Q2. Solve the following.
- A) Explain the Kirchhoff Current Law with a suitable example. 6
 B) Derive an expression for the average and the RMS value in an AC circuit. 6
- Q3. Solve the following.
- A) Distinguished between Motors and Generators with examples. 6
 B) Derive the EMF equation for a transformer with a neatly labelled diagram. 6
- Q4. Solve Any Two of the following.
- A) Explain the principle operation of a PN Junction Diode with a suitable diagram. 6
 B) Explain the working of a full-wave bridge rectifier with waveforms. 6
 C) Explain the functional block diagram of a DC power supply. 6
- Q5. Solve Any Two of the following.
- A) Derive the relation between current gains in Common Base (α) and Common Emitter (β) configurations. 6
 B) Explain the input and output characteristics of a BJT in the Common Emitter (CE) configuration. 6
 C) Enlist Different Transistor Biasing Techniques, and explain the Fixed Biased Technique in detail. 6
- Q6. Solve Any Two of the following.
- A) Explain the block Diagram of a function Generator. 6
 B) Describe the working principle of the Moving Coil instruments with a suitable diagram. 6
 C) Explain the operation of a Digital Storage Oscilloscope with suitable Diagram. 6

*** End ***